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Optical tip-tracking systems and methods thereof

Abstract

An optical tip-tracking system is disclosed including a light-emitting stylet, a light detector, and a console configured to operably connect to the light-emitting stylet and the light detector. The light-emitting stylet can be disposed in a lumen of a catheter. The light-emitting stylet can include a light source in a distal-end portion of the light-emitting stylet configured to emit light. The light detector can be placed over a patient. The light detector can include a plurality of photodetectors configured to detect the light emitted from the light source. The console can be configured to instantiate an optical tip-tracking process for optically tracking the distal-end portion of the light-emitting stylet while the light-emitting stylet is disposed in a vasculature of the patient, the light source is emitting light, the light detector is disposed over the light-emitting stylet, and the photodetectors are detecting the light emitted from the light source.

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Background/Summary

PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/105,310, filed Nov. 25, 2020, now U.S. Pat. No. 11,850,338, which claims the benefit of priority to U.S. Patent Application No. 62/940,107 filed Nov. 25, 2019, each of which is incorporated by reference in its entirety into this application.

BACKGROUND
(1) Intravascular guidance of medical devices such as guidewires and catheters typically requires fluoroscopic methods for tracking tips of the medical devices and determining whether the tips are appropriately localized in their target anatomical structures. Such fluoroscopic methods expose clinicians and patients alike to harmful X-ray radiation. In some cases, the patients are exposed to potentially harmful contrast media needed for the fluoroscopic methods. Magnetic and electromagnetic means for tracking the tips of the medical devices obviate some of the foregoing issues with respect to exposure to radiation and potentially harmful contrast media, but the magnetic and electromagnetic means for tracking the tips of the medical devices are prone to interference.
(2) Disclosed herein are optical tip-tracking systems and methods thereof that address the foregoing.
SUMMARY
(3) Disclosed herein is an optical tip-tracking system including, in some embodiments, a light-emitting stylet, a light detector, and a console configured to operably connect to the light-emitting stylet and the light detector. The light-emitting stylet is configured to be disposed in a lumen of a catheter. The light-emitting stylet includes a light source in a distal-end portion of the light-emitting stylet configured to emit light. The light detector is configured to be placed over a patient. The light detector includes a plurality of photodetectors configured to detect the light emitted from the light source. The console includes memory and a processor configured to instantiate an optical tip-tracking process for optically tracking the distal-end portion of the light-emitting stylet while the light-emitting stylet is disposed in a vasculature of the patient, the light source is emitting light, the light detector is disposed over the light-emitting stylet, and the photodetectors are detecting the light emitted from the light source.
(4) In some embodiments, the light source is a light-emitting diode ("LED").
(5) In some embodiments, the light emitted from the light source has a center wavelength between about 650 nm to 1350 nm.
(6) In some embodiments, the light emitted from the light source has a center wavelength between about 650 nm to 950 nm.
(7) In some embodiments, the distal-end portion of the light-emitting stylet is configured to directionally emit light in one or more chosen directions.
(8) In some embodiments, the light detector includes a housing having a patient-facing portion of the housing configured to transmit at least a portion of the light emitted from the light source to the

photodetectors.
(9) In some embodiments, the housing has a light-blocking portion of the housing opposite the patient-facing portion configured to block ambient light from the photodetectors.
(10) In some embodiments, the photodetectors are arranged in an array such that the light emitted from the light source remains detectable by at least one photodetector of the photodetectors even when the light emitted from the light source is anatomically blocked from another one or more photodetectors of the photodetectors.
(11) In some embodiments, the optical tip-tracking process is configured to provide tracking information as input to a display server of the console for optically tracking the distal-end portion of the light-emitting stylet in a graphical user interface on a display.
(12) In some embodiments, the light-emitting stylet is configured to directly connect to the console.
(13) In some embodiments, the light-emitting stylet is configured to indirectly connect to the console through an intervening multi-use cable.
(14) In some embodiments, the light-emitting stylet is configured to indirectly connect to the console through the light detector.
(15) In some embodiments, the light detector is configured to be placed over the patient and under a sterile drape. The light-emitting stylet includes a drape-piercing connector having a piercing element configured to pierce the sterile drape and insert into a receptacle of a light-detector connector extending from the light detector under the drape.
(16) Also disclosed herein is an optical tip-tracking system including, in some embodiments, a catheter, a light-emitting stylet for the catheter, a light detector, and a console configured to operably connect to the light-emitting stylet and the light detector. The catheter includes a lumen extending through the catheter. The light-emitting stylet is configured to be disposed in a lumen of a catheter. The light-emitting stylet includes a light-emitting diode ("LED") in a distal-end portion of the light-emitting stylet configured to emit light having a center wavelength between about 650 nm to 1350 nm. The light detector is configured to be placed over a patient. The light detector includes a plurality of photodetectors configured to detect the light emitted from the LED. The console includes memory and a processor configured to instantiate an optical tip-tracking process for optically tracking the distal-end portion of the light-emitting stylet in a graphical user interface on a display while the light-emitting stylet is disposed in a vasculature of the patient, the LED is emitting light, the light detector is disposed over the light-emitting stylet, and the photodetectors are detecting the light emitted from the LED.
(17) In some embodiments, the light detector is configured to be placed over the patient and under a sterile drape. The light-emitting stylet includes a drape-piercing connector configured to pierce the sterile drape and connect with a light-detector connector extending from the light detector under the drape.
(18) Also disclosed herein is an optical tip-tracking system including, in some embodiments, a light-emitting stylet, a light detector, and a console configured to operably connect to the light-emitting stylet and the light detector. The light-emitting stylet is configured to be disposed in a lumen of a catheter. The light-emitting stylet includes an optical fiber configured to convey light to a distal-end portion of the light-emitting stylet for emitting light therefrom. The light detector is configured to be placed over a patient. The light detector includes a plurality of photodetectors configured to detect the light emitted from the light source. The console includes a light source for the light emitting stylet and memory and a processor. The memory and the processor are configured to instantiate an optical tip-tracking process for optically tracking the distal-end portion of the light-emitting stylet while the light-emitting stylet is disposed in a vasculature of the patient, the light source is emitting light, the light detector is disposed over the light-emitting stylet, and the photodetectors are detecting the light emitted from the light source.
(19) Also disclosed herein is a method of an optical tip-tracking system including, in some embodiments, a disposing step of disposing a light-emitting stylet of the optical tip-tracking system

in a lumen of a catheter. The light-emitting stylet includes a light source in a distal-end portion of the light-emitting stylet. The method also includes a placing step of placing a light detector of the optical tip-tracking system over a patient. The light detector includes a plurality of photodetectors. The method also includes an advancing step of advancing the catheter from an insertion site to a destination within a vasculature of the patient while emitting light from the light source and detecting the light with the photodetectors. The method also includes a viewing step of viewing a display screen of the optical tip-tracking system while the display screen graphically tracks the distal-end portion of the light-emitting stylet through the vasculature of the patient.

(20) In some embodiments, the light source of the light-emitting stylet distally extends beyond a distal end of the catheter while advancing the catheter, thereby enabling the photodetectors of the light detector to detect the light emitted from the light source.

(21) In some embodiments, the method also includes a placing step of placing a sterile drape over both the patient and the light detector. The method also includes a connecting step of connecting a drape-piercing connector of the light-emitting stylet with a light-detector connector extending from the light detector. The connecting step includes piercing the sterile drape with a piercing element of the drape-piercing connector before inserting the piercing element into a receptacle of the light-detector connector.

(22) In some embodiments, the catheter is a central venous catheter (“CVC”). The advancing step includes advancing the CVC with the light-emitting stylet disposed therein through a right internal jugular vein, a right brachiocephalic vein, and into a superior vena cava (“SVC”).

(23) In some embodiments, the catheter is a peripherally inserted central catheter (“PICC”). The advancing step includes advancing the PICC with the light-emitting stylet disposed therein through a right basilic vein, a right axillary vein, a right subclavian vein, a right brachiocephalic vein, and into an SVC.

(24) In some embodiments, the method also includes a ceasing step of ceasing to advance the catheter through the vasculature of the patient after determining the distal-end portion of light-emitting stylet is located at the destination by way of the display screen.

(25) These and other features of the concepts provided herein will become more apparent to those of skill in the art in view of the accompanying drawings and following description, which describe particular embodiments of such concepts in greater detail.

Description

DRAWINGS

(1) FIG. **1** provides a block diagram of a first optical tip-tracking system in accordance with some embodiments.

(2) FIG. **2** provides a block diagram of a second optical tip-tracking system in accordance with some embodiments.

(3) FIG. **3** illustrates the first optical tip-tracking system in accordance with some embodiments.

(4) FIG. **4** illustrates the first optical tip-tracking system including a catheter in accordance with some embodiments.

(5) FIG. **5**A illustrates a distal-end portion of a first light-emitting stylet of the first or second optical tip-tracking system in accordance with some embodiments.

(6) FIG. **5**B illustrates a transverse cross section of the first light-emitting stylet disposed in the catheter in accordance with some embodiments.

(7) FIG. **6**A illustrates a distal-end portion of a second light-emitting stylet of the first or second optical tip-tracking system in accordance with some embodiments.

(8) FIG. **6**B illustrates a transverse cross section of the second light-emitting stylet disposed in the catheter in accordance with some embodiments.

(9) FIG. **7** illustrates a side view of a light detector of the first or second optical tip-tracking system in accordance with some embodiments.
(10) FIG. **8** illustrate the first optical tip-tracking system in use in accordance with some embodiments.
(11) FIG. **9** illustrate the second optical tip-tracking system in use in accordance with some embodiments.
(12) FIG. **10** provides a transmission curve for optical window for biological tissue in accordance with some embodiments.

DESCRIPTION

(13) Before some particular embodiments are disclosed in greater detail, it should be understood that the particular embodiments disclosed herein do not limit the scope of the concepts provided herein. It should also be understood that a particular embodiment disclosed herein can have features that can be readily separated from the particular embodiment and optionally combined with or substituted for features of any of a number of other embodiments disclosed herein.
(14) Regarding terms used herein, it should also be understood the terms are for the purpose of describing some particular embodiments, and the terms do not limit the scope of the concepts provided herein. Ordinal numbers (e.g., first, second, third, etc.) are generally used to distinguish or identify different features or steps in a group of features or steps, and do not supply a serial or numerical limitation. For example, “first,” “second,” and “third” features or steps need not necessarily appear in that order, and the particular embodiments including such features or steps need not necessarily be limited to the three features or steps. Labels such as “left,” “right,” “top,” “bottom,” “front,” “back,” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. Singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
(15) With respect to “proximal,” a “proximal portion” or a “proximal-end portion” of, for example, a catheter disclosed herein includes a portion of the catheter intended to be near a clinician when the catheter is used on a patient. Likewise, a “proximal length” of, for example, the catheter includes a length of the catheter intended to be near the clinician when the catheter is used on the patient. A “proximal end” of, for example, the catheter includes an end of the catheter intended to be near the clinician when the catheter is used on the patient. The proximal portion, the proximal-end portion, or the proximal length of the catheter can include the proximal end of the catheter; however, the proximal portion, the proximal-end portion, or the proximal length of the catheter need not include the proximal end of the catheter. That is, unless context suggests otherwise, the proximal portion, the proximal-end portion, or the proximal length of the catheter is not a terminal portion or terminal length of the catheter.
(16) With respect to “distal,” a “distal portion” or a “distal-end portion” of, for example, a catheter disclosed herein includes a portion of the catheter intended to be near or in a patient when the catheter is used on the patient. Likewise, a “distal length” of, for example, the catheter includes a length of the catheter intended to be near or in the patient when the catheter is used on the patient. A “distal end” of, for example, the catheter includes an end of the catheter intended to be near or in the patient when the catheter is used on the patient. The distal portion, the distal-end portion, or the distal length of the catheter can include the distal end of the catheter; however, the distal portion, the distal-end portion, or the distal length of the catheter need not include the distal end of the catheter. That is, unless context suggests otherwise, the distal portion, the distal-end portion, or the distal length of the catheter is not a terminal portion or terminal length of the catheter.
(17) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art.
(18) Again, fluoroscopic methods typically used for tracking tips of medical devices such as guidewires and catheters expose clinicians and patients alike to harmful X-ray radiation. Magnetic

and electromagnetic means for tracking the tips of the medical devices obviate some of the foregoing issues with respect to exposure to radiation, but the magnetic and electromagnetic means for tracking the tips of the medical devices are prone to interference.

(19) Disclosed herein are optical tip-tracking systems and methods thereof that address the foregoing.

(20) Optical Tip-Tracking Systems

(21) FIG. **1** provides a block diagram of a first optical tip-tracking system **100**, FIG. **3** illustrates the first optical tip-tracking system **100**, FIG. **4** illustrates the first optical tip-tracking system **100** including a catheter **450**, and FIG. **8** illustrates the first optical tip-tracking system **100** in use, in accordance with some embodiments. FIG. **2** provides a block diagram of a second optical tip-tracking system **200** and FIG. **9** illustrates the second optical tip-tracking system **200** in use, in accordance with some embodiments.

(22) As shown, the optical tip-tracking system **100** or **200** includes a light-emitting stylet **110**, a light detector **120**, and a console **130** or **230** configured to operably connect to the light-emitting stylet **110** and the light detector **120**. Each optical tip-tracking system of the optical tip-tracking systems **100** and **200** also includes a display screen; however, the optical tip-tracking system **100** includes a standalone display screen **140**, whereas the optical tip-tracking system **200** includes an integrated display screen **240**. Each optical tip-tracking system of the optical tip-tracking systems **100** and **200** can also include a medical device **150** such as the catheter **450** of FIG. **4**.

(23) Beginning with the consoles **130** and **230**, the console **130** or **230** includes memory **134** such as primary memory **136** and secondary memory **138**. The primary memory **136** includes random-access memory (“RAM”). The secondary memory **138** includes non-volatile memory such as read-only memory (“ROM”) including a set of instructions **139** or **239** for loading into the primary memory **136** at runtime of the console **130** or **230**.

(24) One or more processors **132** are configured to instantiate an optical tip-tracking process in accordance with the instructions **139** or **239** for optically tracking a distal-end portion of the light-emitting stylet **110** while the light-emitting stylet **110** is disposed in a vasculature of a patient, a light source such as one or more LEDs **112** is emitting light, the light detector **120** is disposed over the light-emitting stylet **110**, and a plurality of photodetectors of the light detector **120** such as photodetectors **122** are detecting the light emitted from the light source of the light-emitting stylet **110**. The optical tip-tracking process is configured to provide tracking information as input to a display server of the console **130** or **230** for optically tracking the distal-end portion of the light-emitting stylet **110** in a graphical user interface on the display screen **140** or **240**. Optically tracking the distal-end portion of the light-emitting stylet **110** can be animated on the display screen **140** or **240** as shown on the display screen **240** of the console **230** in FIGS. **8** and **9**.

(25) FIGS. **1** and **2** provide different connection options for connecting at least the light-emitting stylet **110** and the light detector **120** to the console **130** or **230**.

(26) The left-hand side of each figure or FIGS. **1** and **2** shows a connection option (“option A”) in which the light-emitting stylet **110** and the light detector **120** are independently connected to the console **130** or **230**. For example, the console **130** or **230** and the light-emitting stylet **110** can be mutually configured such that the light-emitting stylet **110** directly connects to the console **130** or **230** or indirectly connects to the console **130** or **230** through an intervening multi-use cable having requisite conveying means for conveying electrical power (e.g., electrical leads), light (e.g., optical fiber), etc. Likewise, the console **130** or **230** and the light detector **120** can be mutually configured such that the light detector **120** directly connects to the console **130** or **230** or indirectly connects to the console **130** or **230** through an intervening multi-use cable.

(27) The right-hand side of each figure of FIGS. **1** and **2** shows a connection option (“option B”) in which the light-emitting stylet **110** is connected to the light detector **120** and, in turn, the light detector **120** is connected to the console **130** or **230**. A cable such as the intervening multi-use cable set forth above for connecting the light detector **120** to the console **130** or **230** includes the requisite

conveying means for conveying electrical power (e.g., electrical leads), light (e.g., optical fiber), etc. The foregoing connection option is useful in that the light detector **120** is multi-use equipment configured to be placed over a patient P and under a sterile drape **801** outside a sterile field as shown in FIGS. **8** and **9**. The light-emitting stylet **110** is single-use equipment intended for use within the foregoing sterile field. As shown in FIG. **9**, the light-emitting stylet **110** can include a drape-piercing connector **900** in a proximal-end portion thereof having a piercing element **902** configured to both pierce the sterile drape **801** and insert into a receptacle **904** of a light-detector connector extending from the light detector **120** under the sterile drape **801**. So configured, the intervening multi-use cable set forth above through which the light-emitting stylet **110** can be indirectly connected to the console **130** or **230** is not needed. For examples of drape-piercing connectors for the light-emitting stylet **110** and light-detector connectors for the light-detector **120** see U.S. Pat. No. 10,231,753 and children patents or patent applications thereof, each of which patents and patent application are incorporated herein by reference.
(28) Both connection options of FIG. **1** show the display screen **140** connected to the console **130** in the optical tip-tracking system **100**. FIG. **2** does not show connection options for connecting the integrated display screen **240** to the console **230** in the optical tip-tracking system **200**. This is because the integrated display screen **240** is integrated into the console **240** as shown in FIGS. **3**, **4**, **8**, and **9**.
(29) In addition to FIGS. **1-4**, FIG. **7** illustrates a side view of the light detector **120** in accordance with some embodiments.
(30) As set forth above, the light detector **120** is configured to be placed over a patient and under a sterile drape such as the sterile drape **801** of FIGS. **8** and **9**.
(31) The light detector **120** includes the photodetectors **122** disposed within the light detector **120** configured to detect the light emitted from the light source of the light-emitting stylet **110**. The photodetectors **122** are arranged in an array such that the light emitted from the light source of the light-emitting stylet **110** remains detectable by at least one photodetector of the photodetectors **122** even when the light emitted from the light source of the light-emitting stylet **110** is anatomically blocked (e.g. by a rib) from another one or more photodetectors of the photodetectors **122**.
(32) The light detector **120** includes a housing having a patient-facing portion **722** of the housing configured to transmit at least a portion of the light emitted from the light source of the light-emitting stylet **110** to the photodetectors **122** disposed within the light detector **120**. The housing also has a light-blocking portion **724** of the housing opposite the patient-facing portion **722** of the housing configured to block ambient light from the photodetectors **122** disposed within the light detector **120**. In addition to the light-blocking portion **724** of the housing of the light detector **120**, the sterile drape **801** also protects the photodetectors **122** from the ambient light while in use.
(33) In addition to FIGS. **1-4**, FIG. **5**A illustrates the distal-end portion of the light-emitting stylet **110** and FIG. **5**B illustrates a transverse cross section of the light-emitting stylet **110** in accordance with some embodiments having the one-or-more LEDs **112** as the light source. FIG. **6**A illustrates the distal-end portion of the light-emitting stylet **110** and FIG. **6**B illustrates a transverse cross section of the light-emitting stylet **110** in accordance with some embodiments having an external light source.
(34) As set forth in more detail below, the light-emitting stylet **110** is configured to be disposed in a lumen of a catheter.
(35) With respect to the light-emitting stylet **110** having the one-or-more LEDs **112** as the light source, the light-emitting stylet **110** includes the one-or-more LEDs **112** in a distal-end portion (e.g., a tip) of the light-emitting stylet **110** configured to emit light. The light-emitting stylet **110** also includes at least a pair of electrical leads **114** configured to convey electrical power from the console **130** or **230** to power the one-or-more LEDs **112**. The distal-end portion of the light-emitting stylet **110** can be configured to directionally emit light from the one-or-more LEDs **112** in one or more chosen directions such as straight ahead in line with the light-emitting stylet **110**,

radially outward such as toward an extracorporeal surface of a patient, or a combination thereof.
(36) With respect to the light-emitting stylet **110** having the external light source, the light-emitting stylet **110** includes an optical fiber **615** configured to convey light from the external source, for example, a light within the console **130** or **230**, to the distal-end portion (e.g., the tip) of the light-emitting stylet **110** to emit light. A ferrule **613** disposed over a distal-end portion of the optical fiber **615** can be configured to directionally emit light from the optical fiber **615** in one or more chosen directions such as straight ahead in line with the light-emitting stylet **110**, radially outward such as toward an extracorporeal surface of a patient, or a combination thereof.
(37) The light emitted from the light source can have a center wavelength between about 650 nm to 1350 nm including a center wavelength between about 650 nm to 950 nm or a center wavelength between about 1100 nm to 1350 nm. The light in the foregoing ranges of wavelengths is within an optical window for biological tissue in that such light penetrates biological tissue more deeply than light outside the foregoing ranges of wavelengths. (See FIG. **10** for the optical window for biological tissue.) Each LED of the one-or-more LEDs **112** can include aluminum gallium arsenide (AlGaAs) or gallium arsenide (GaAs) as a semiconductor material. In accordance with the semiconductor material and its composition, each LED of the one-or-more LEDs **112** can be configured to emit light at or within the following wavelengths: 660 nm, 680 nm, 800-850 nm, 850-940 nm, and 940 nm.
(38) Adverting to FIG. **4**, the catheter **450** can be a peripherally inserted central catheter ("PICC"), as shown, or a central venous catheter ("CVC"). In an example of a diluminal catheter, the catheter **450** can include a catheter tube **452**, a bifurcated hub **454**, two extension legs **456**, and two Luer connectors **458** operably connected in the foregoing order. Notwithstanding the foregoing example, the catheter **450** can alternatively be a monoluminal catheter or a multi-luminal catheter including three or more lumens.
(39) Continuing with the example of the diluminal catheter, the catheter **450** includes two lumens extending therethrough formed of adjoining lumen portions. Indeed, the catheter tube **452** includes two catheter-tube lumens **553**. (See FIGS. **5**B and **6**B for the two catheter-tube lumens **553**). The bifurcated hub **454** has two hub lumens correspondingly fluidly connected to the two catheter-tube lumens **553**. Each extension leg of the two extension legs **456** has an extension-leg lumen fluidly connected to a hub lumen of the two hub lumens. Either lumen extending through the catheter **450** can accommodate the light-emitting stylet **110** disposed therein.
(40) Methods
(41) A method of the optical tip-tracking system **100** or **200** includes a disposing step of disposing the light-emitting stylet **110** of the optical tip-tracking system **100** or **200** in a lumen of the catheter **450**.
(42) The method also includes a placing step of placing the light detector **120** of the optical tip-tracking system **100** or **200** over the patient P as shown in FIGS. **8** and **9**.
(43) The method also includes a placing step of placing the sterile drape **801** over both the patient P and the light detector **120**.
(44) As shown in FIG. **8**, the method can also include a connecting step of connecting the drape-piercing connector of the light-emitting stylet **110** with the light-detector connector extending from the light detector **120**. Such a connecting step includes piercing the sterile drape **801** with the piercing element of the drape-piercing connector before inserting the piercing element into the receptacle of the light-detector connector.
(45) The method also includes an advancing step of advancing the catheter **450** from an insertion site to a destination within a vasculature of the patient P while emitting light from the light source (e.g., the one-or-more LEDs **112**) and detecting the light with the photodetectors **122**. The light source of the light-emitting stylet **110** should distally extends beyond a distal end of the catheter **450** while advancing the catheter **450**, thereby enabling the photodetectors **122** of the light detector **120** to detect the light emitted from the light source of the light-emitting stylet **110**.

(46) When the catheter **450** is a CVC, the advancing step includes advancing the CVC with the light-emitting stylet **110** disposed therein through a right internal jugular vein, a right brachiocephalic vein, and into an SVC.
(47) When the catheter **450** is a PICC, the advancing step includes advancing the PICC with the light-emitting stylet **110** disposed therein through a right basilic vein, a right axillary vein, a right subclavian vein, a right brachiocephalic vein, and into an SVC.
(48) The method also includes a viewing step of viewing the display screen **140** or **240** of the optical tip-tracking system **100** or **200** while the display screen **140** or **240** graphically tracks the distal-end portion of the light-emitting stylet **110** through the vasculature of the patient P.
(49) The method also includes a ceasing step of ceasing to advance the catheter **450** through the vasculature of the patient P after determining the distal-end portion of the light-emitting stylet **110** is located at the destination by way of the display screen **140** or **240**.
(50) While some particular embodiments have been disclosed herein, and while the particular embodiments have been disclosed in some detail, it is not the intention for the particular embodiments to limit the scope of the concepts provided herein. Additional adaptations and/or modifications can appear to those of ordinary skill in the art, and, in broader aspects, these adaptations and/or modifications are encompassed as well. Accordingly, departures may be made from the particular embodiments disclosed herein without departing from the scope of the concepts provided herein.

Claims

1. A medical device tip-tracking system, comprising: a light-emitting stylet configured to be disposed in a lumen of a medical device, the light-emitting stylet configured to emit light from a distal-end portion of the light-emitting stylet; a light detector configured to be placed over and external to a patient, the light detector including a plurality of photodetectors configured to detect the light emitted from a light source, wherein the plurality of photodetectors are arranged in the light detector such that the plurality of photodetectors are configured to detect a remaining portion of the light when a first portion of the light is anatomically blocked by a bone of the patient; and a console coupled to the light-emitting stylet and the light detector, the console including: a processor, and non-volatile memory storing instructions that, when executed by the processor, cause performance of operations including: displaying tracking information in a graphical user interface on a display screen as the light-emitting stylet is advanced through a vasculature of the patient.
2. The medical device tip-tracking system of claim 1, wherein the light-emitting stylet is configured to directly connect to the light detector.
3. The medical device tip-tracking system of claim 1, wherein the light-emitting stylet is configured to directly connect to the console.
4. The medical device tip-tracking system of claim 1, wherein the light-emitting stylet includes the light source at the distal-end portion, wherein the light source is configured to emit the light.
5. The medical device tip-tracking system of claim 4, wherein the light-emitting stylet is configured to indirectly connect to the console through the light detector such that electrical power passes from either the console to the light source or the light detector.
6. The medical device tip-tracking system of claim 1, wherein the light-emitting stylet includes an electrical lead configured to convey electrical power to the light source.
7. The medical device tip-tracking system of claim 1, wherein the light-emitting stylet includes an optical fiber configured to convey the light along a length of the light-emitting stylet, wherein the light is emitted from a distal-end portion of the optical fiber.
8. The medical device tip-tracking system of claim 7, wherein the light-emitting stylet includes a ferrule disposed over the distal-end portion of the optical fiber, and wherein the ferrule is

configured to direct the light in one or more directions.
9. The medical device tip-tracking system of claim 1, wherein the instructions, when executed by the processor, cause performance of further operations including: graphically tracking the distal-end portion of the light-emitting stylet through the vasculature of the patient.
10. The medical device tip-tracking system of claim 1, wherein the console further includes the display screen integrated into a housing of the console.
11. A method of deploying an optical tip-tracking system, comprising: disposing a light-emitting stylet in a lumen of a medical device, the light-emitting stylet configured to emit light from a distal-end portion of the light-emitting stylet; placing a light detector over and external to a patient, the light detector including a plurality of photodetectors configured to detect the light emitted from a light source, wherein the plurality of photodetectors are arranged in the light detector such that the plurality of photodetectors are configured to detect a remaining portion of the light when a first portion of the light is anatomically blocked by a bone of the patient; advancing the light-emitting stylet through a vasculature of the patient; and viewing tracking information on a graphical user interface on a display screen as the light-emitting stylet is advanced through the vasculature of the patient.
12. The method of claim 11, wherein the light-emitting stylet is configured to directly connect to the light detector.
13. The method of claim 11, wherein the light-emitting stylet is configured to directly connect to a console.
14. The method of claim 11, wherein the light-emitting stylet includes the light source at the distal-end portion, wherein the light source is configured to emit the light.
15. The method of claim 14, wherein the light-emitting stylet is configured to indirectly connect to a console through the light detector such that electrical power passes from either the console to the light source or to the light detector.
16. The method of claim 11, wherein the light-emitting stylet includes an electrical lead configured to convey electrical power to the light source.
17. The method of claim 11, wherein the light-emitting stylet includes an optical fiber configured to convey the light along a length of the light-emitting stylet, wherein the light is emitted from a distal-end portion of the optical fiber.
18. The method of claim 17, wherein the light-emitting stylet includes a ferrule disposed over the distal-end portion of the optical fiber, and wherein the ferrule is configured to direct the light in one or more directions.
19. The method of claim 11, wherein instructions, when executed by a processor, cause performance of further operations including: graphically tracking the distal-end portion of the light-emitting stylet through the vasculature of the patient.
20. The method of claim 11, wherein a console further includes the display screen integrated into a housing of the console.